

The World-Wide Web As a Data Source

Topics

- HTML Basics
- Scraping Basics
- Advanced Web Scraping

HTML Code

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Welcome to CS 301</title>
5  </head>
6  <body>
7      <H1>Welcome to CS 301 – Introduction to Data Science</H1>
8      <p style="color:blue">Today we will learn how to gather
9          data from the web.
10     </p>
11     <p id="html_paragraph">But first we need to learn a little
12         about HTML.
13     </p>
14 </body>
15 </html>
```

Browser

Welcome to CS 301 - Introduction to Data Science

Today we will learn how to gather data from the web.

But first we need to learn a little about HTML.

TAG **ATTRIBUTE** **TEXT**

<p id="html_paragraph">But first we need to learn a little about HTML.</p>

HTML Element

```
<tag-name attr1="attr1 value" attr2="attr2 value" ... >  
    Inner HTML text of this tag  
</tag-name>
```

Common Tags

Tag	Use
h1, h2, h3, ...	Headers
p	Paragraph
a	Hyperlink (anchor)
div	Section of a page
img	Image
li	List item
tr	Table row
td	Table column
br	New line

Common Attributes

Attribute	Use
id	Unique identifier
class	Style class
style	Extra styling
href	Linked URL
src	Source file

Document Object Model

HTML

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Welcome to CS 301</title>
5 </head>
6 <body>
7   <div id="header_div">
8     <H1>Welcome to CS 301 - Introduction to Data
9     Science</H1>
10  </div>
11  <div id="message_div">
12    <p style="color:blue">Today we will learn how to gather
13    data from the web.</p>
14    <p id="html_paragraph">But first we need to learn a
15    little about HTML.</p>
16    <a href="https://www.w3.org/standards/">
17      WWW Standards</a>
18  </div>
19 </body>
20 </html>
21
```

DOM

